MILITARY SPECIFICATION
RECEIVING SET, RADIO AN/DRW-29

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall receive and decode UHF FM radio control signals in the frequency range of 406 to 549.5 megacycles having modulation deviation of plus and minus 120 kilocycles, and without having serious malfunction caused by deviation up to plus and minus 300 kilocycles. The equipment shall be intended for installation in pilotless aircraft or missiles wherein the output functions of this equipment shall be utilized to activate various electromechanical control devices to control various functions within the aircraft or missile.

1.2 Classification - The Receiving Set, Radio AN/DRW-29, shall consist of the following items:

Items	Type Designation	Applicable Paragraph
Receiver, Radio	R-1136(*)/DRW-29	3.4.1
Base, Mounting	MT-2740(*)/DRW-29	3.4.2

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-C-172	Cases, Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-C-3608	Connector, "BNC" for Radio Frequency Cables

Source: https://assist.dla.mil -- Downloaded: 2026-01-19T17:26Z
Check the source to verify that this is the current version before use.

Military

MIL-E-4682	Electron Tubes and Transistor, Choice and Application of
MIL-W-5088	Wiring; Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of Repair Parts
MIL-T-18303	Test Procedures; Preproduction and Inspection For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-C-26482	Connectors, Electric, "MS" Type

STANDARDS

Military

MIL-STD-415	Test Points and Test Facilities, Design Standards for
MIL-STD-439	Electronic Circuits
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MS 25245	Connections, Input Power Airborne Electronic Equipment; 28-volt DC, 115-volt Single Phase AC, and/or 115/200-volt Three Phase AC, or combinations

Source: https://assist.dla.mil - Downloaded: 202640-19T17:26Z
Check the source to verify that this is the current version before use.

2.2 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400 except where otherwise dictated by size, weight, or performance considerations.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

3.1.2 Electron Devices - Electron tubes, transistors, and diodes shall be chosen and applied, and the complements reported as outlined in Specification MIL-E-4682. The complement report must be submitted to the procuring activity for review and approval prior to preproduction testing.

3.2 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, unless otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the current version before use.

3.2.1 Total Weight - The total weight of the equipment shall not exceed 11 pounds.

3.2.2 Operating Life -

3.2.2.1 Operational Stability - The equipment shall operate with optimum performance for 50 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.2.2 Reliable Operating Life - The equipment shall have a reliable operating life of at least 100 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval, in order to renew this service life, shall be specified by the manufacturer.

3.2.2.3 Total Operating Life - The equipment shall have a minimum total operating life of 500 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.3 Cabling and Connections -

3.2.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400, except that the connectors shall be miniature type connectors in accordance with MIL-C-26482 where applicable. RF connectors shall be in accordance with MIL-C-3608.

3.2.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

3.2.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

3.2.5 Interference Control - The generation of interference by the equipment and vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.6 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.

3.2.7 Nomenclature and Nameplates - Nomenclature assigments and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.8 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (30° ±10° C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	27.5 ±0.5 V DC

3.2.9 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified herein.

3.2.9.1 Vibration - The equipment shall operate satisfactorily when subjected to the vibration requirements of Curves I and II of Specification MIL-E-5400.

3.2.9.2 Temperature - The equipment shall withstand a maximum temperature of +71° C under operating conditions and a maximum temperature of +85° C under storage conditions.

3.2.10 Warmup Period - The time required for the equipment to warm up prior to operation shall be kept to a minimum and shall not exceed 1.0 minute under all service conditions.

3.2.11 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

(1) DC Power 28 V Category "B" 0.4 Amps.

3.2.11.1 Input Power Connection - Input power connections shall be in accordance with MS 25245

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the currentversion before use.

3.2.11.2 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.3 Performance - Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under both standard and extreme service conditions. When reduced performance under the extreme service conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under standard conditions will be specified. Sensitivity figures assume a 50 ohm resistive source impedance (antenna system or signal generator).

3.3.1 Operation - The Receiving Set, Radio AN/DRW-29, shall be capable of receiving and decoding properly modulated command signals transmitted by a distant UHF FM radio transmitter which provide an input signal amplitude of five microvolts or more to the RF input connector of the Receiver, Radio R-1136(*)/DRW-29. The decoded command signals shall be made available as relay contact closures and openings in accordance with the tabulation shown in 3.3.8. Twenty decoder channels shall be provided. Any number of channels from one to six inclusive may be operated without degradation of specified performance when the peak deviation does not exceed 300 kc for wide band operation or 120 kc for narrow band operation (plus and minus), and the deviation for any specific channel exceeds 45 kc (50 kc nominal) for wide band and 18 kc (20 kc nominal) for narrow band operation. A carrier operated relay circuit shall be incorporated in the equipment. This circuit shall provide an externally available set of single pole, double throw relay contacts. The relay shall operate to its normal, fail-safe position at any time that the Receiver, Radio R-1136(*)/DRW-29 receives an RF input signal of less than five microvolts or whenever power is supplied to the Receiver, Radio R-1136(*)/DRW-29, is interrupted. The relay shall operate to its energized position whenever the Receiver, Radio R-1136(*)/DRW-29 receives an RF input in excess of some value between five and ten microvolts. Under standard conditions, the carrier relay shall not require an input to the receiver in excess of 10 microvolts to become energized.

3.3.1.1 Wide/Narrow Band Switch - In order to provide maximum performance for both the wide band (300 kc deviation) and narrow band (120 kc deviation) operation, an internal switch shall be provided. The switch position shall be labeled "300 kc" and "120 kc". Adjacent to the channel indicator block on the front panel will be a block labeled "Peak Deviation", on which shall be marked either "300 kc" or "120 kc" to correspond to the position of the internal selector switch. With the switch in the "120 kc" position, the receiver shall handle unintentional excess deviation up to plus and minus 150 kc (total) without degradation of the performance characteristics.

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the current version before use.

3.3.2 Frequency Characteristics -

3.3.2.1 Radio Frequency Characteristics - The receiver shall operate at any desired frequency between 406 and 549.5 megacycles. The actual operating frequency shall be controlled by a piezoelectric quartz crystal. Each crystal shall provide for operation at either of two frequencies, one 33.5 megacycles above and one 33.5 megacycles below the local oscillator frequency, depending upon the tuning of the receiver preselector section. Specific crystal frequencies shall be derived utilizing the following formula:

$$F_{xtal} = \frac{F_O \pm 33.5}{9}$$

when: F_{xtal} = crystal frequency in megacycles

and: F_O = operating frequency in megacycles

3.3.2.2 Audio Frequency Characteristics - The audio frequency circuits shall provide operation of the decoder portion of the equipment for the following frequencies:

Channel 1	7,500 cps	Channel 11	25,010 cps
Channel 2	8,460 cps	Channel 12	28,210 cps
Channel 3	9,540 cps	Channel 13	31,830 cps
Channel 4	10,760 cps	Channel 14	35,900 cps
Channel 5	12,140 cps	Channel 15	40,490 cps
Channel 6	13,700 cps	Channel 16	45,680 cps
Channel 7	15,450 cps	Channel 17	51,520 cps
Channel 8	17,430 cps	Channel 18	58,120 cps
Channel 9	19,660 cps	Channel 19	65,560 cps
Channel 10	22,170 cps	Channel 20	73,950 cps

Up to any six selected audio channels shall be capable of operating simultaneously without any degradation of operation characteristics in accordance with paragraphs 3.3 and 3.3.1.1.

3.3.3 Tolerances - Tolerances of frequency response of radio frequency channels shall be ±0.01 percent. Tolerances of audio channel frequency response shall be ±1.0 percent.

3.3.4 Sensitivity - The equipment shall provide reliable reception of distant signals providing an input amplitude within the range of 5 microvolts to 2 volts under standard conditions and within the range of 20 microvolts to 2 volts under extreme environmental conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the currentversion before use.

3.3.5 RF Bandwidth - The RF Bandwidth of the equipment shall provide a 3 db passband which will accommodate 300 kc peak deviation of tones up to 73,950 cycles, together with the most unfavorable addition of maximum tolerances on transmitter frequency, receiver crystal, and IF center frequency drift, without producing excessive distortion or other effects which preclude meeting performance specifications set forth herein. The passband at 60 db down shall be less than 3.0 mc.

3.3.6 Image and Spurious Response - Response of image, spurious and parasitic frequency signals shall be at least 60 db less than response to the desired signal.

3.3.7 Command Decoding Capability - The equipment shall be capable of separating commands received in any combination of from one to six of the audio frequencies specified in paragraph 3.3.2.2. Normal functions of the equipment shall occur by reason of any combination of the frequencies specified when from one to six channels are activated simultaneously.

3.3.8 Command Channel Relay Outputs - Each command channel shall operate a double pole, double throw relay. Individual outputs in accordance with the following table shall be available at a connector pin on the front panel. For the purposes of this tabulation each relay is considered to have an "A" section and a "B" section. Each section is comprised of a normally open, NO, contact; a normally closed, NC, contact; and a center arm, C, contact. The contacts are available at the front panel as tabulated below. Each pair of relay contacts shall be capable of conducting 2 amperes at 32 volts DC resistive.

CHANNEL NO.	CONTACT	RELAY SECTION
1 through 5 and	NO	A
11 through 15	NC	A
(Identical)	C	A
	NO	B
	C	B
6 through 10 and	NO	A
16 through 20	NC	A
(Identical)	C*	A

*Center arms for channels 6 through 10 shall be commonly connected and center arms for channels 16 through 20 shall be commonly connected.

3.3.9 Carrier Operated Relay - The carrier operated relay shall be a double pole, double throw relay which operates in a fail-safe type of circuit so that loss of power or reduction of carrier strength below five

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the current version before use.

microvolts will cause the relay to become deenergized. One set of relay contacts comprising a normally open, a normally closed, and a center contact shall be available in an ungrounded circuit at a front panel connector. The second set of contacts shall serve to interrupt the mixed audio to the decoder sections when the carrier operated relay becomes deenergized. The relay shall have a maximum pull-in requirement of not more than ten microvolts RF carrier input to the BNC front panel connector. The relay shall drop out at not less than five microvolts RF carrier input to the BNC front panel connector. These levels of operation are specified at standard conditions. A degradation of not more than four to one may occur at extreme service conditions.

3.4 Detail Requirements - The equipment shall meet the following requirements.

3.4.1 Receiver, Radio R-1136(*)/DRW-29 -

3.4.1.1 Function - The equipment shall serve to provide capabilities of receiving and decoding radio frequency command signals transmitted from remote UHF FM control transmitters. The equipment shall provide relay contact outputs corresponding to the operation of from one to six simultaneous ON-OFF type commands.

3.4.1.2 Form Factor - The maximum external dimensions of the receiver shall be 5-13/32 by 6-29/32 by 8 inches.

3.4.1.3 Weight - The weight of the receiver shall not exceed 9.5 pounds.

3.4.1.4 Contents of Unit - The receiver shall utilize subassembly type construction. The following assemblies shall be provided.

(1) Main Chassis Assembly

(2) Receiver-Subassembly

(3) Decoder-Subassemblies (5)

3.4.1.4.1 Main Chassis Assembly - The main chassis assembly shall include the front panel, carrier operated relay circuitry and mounting of all relays. Provisions for accommodating the five decoder subassemblies and the receiver subassembly shall also be incorporated in the main chassis assembly.

3.4.1.4.1.1 Front Panel - The front panel shall include receptacles for connection of the receiver to external primary power, antenna and external control circuitry.

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the currentversion before use.

3.4.1.4.1.2 Carrier Operated Relay Circuit - The carrier operated relay circuit shall include the direct coupled amplifier circuitry necessary to operate the carrier operated relay. The amplifier portions shall be controlled by a detected sample of the signal present at the last IF amplifier stage of the receiver.

3.4.1.4.2 Receiver Assembly - The receiver assembly shall include a local oscillator, a preselector, a mixer, an intermediate frequency amplifier, an FM detector, an audio amplifier, and a voltage regulator.

3.4.1.4.2.1 Local Oscillator - The local oscillator section shall provide a transistorized crystal controlled oscillator circuit, a transistorized frequency tripler circuit and a diode type frequency tripler circuit. The local oscillator section shall provide a signal which is frequency stabilized to within ±0.01 percent of the desired frequency. Crystal frequencies shall be as specified in paragraph 3.3.2.1.

3.4.1.4.2.2 Preselector - The preselector section shall consist of three tuned-line cavity circuits having resonant frequencies tuneable over the operating frequency range of the equipment.

3.4.1.4.2.3 Mixer - The mixer shall consist of a nonlinear diode type mixer circuit which shall mix the output signal of the preselector section with the output signal of the local oscillator section to produce the difference or intermediate frequency signal. The frequency of the preselector section output signal shall be either above or below the local oscillator frequency depending on the tuning of the preselector section.

3.4.1.4.2.4 Intermediate Frequency Amplifier - The intermediate frequency amplifier shall consist of eight tuned-frequency amplifier stages. The first four of these stages shall be double tuned and the remaining four shall be single tuned. The last intermediate frequency amplifier stage shall be utilized as a driver stage for the FM detector stage. Circuitry capable of producing an output signal proportional in amplitude to the amplitude of the signal present at the last intermediate frequency amplifier stage shall be included. The output signal fromthis circuitry shall be utilized to control the carrier controlled relay circuitry of the main chassis assembly. The intermediate frequency shall be 33.5 megacycles.

3.4.1.4.2.5 FM Detector - The FM detector shall utilize an unbalanced ratio detector type circuit.

3.4.1.4.2.6 Audio Amplifier - The audio amplifier section shall consist of a two stage, transformer coupled transistorized audio amplifier. The output stage shall utilize two transistors in push-pull configuration. The audio amplifier shall provide sufficient output amplitude to drive the decoder.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.

3.4.1.4.2.7 Voltage Regulator - The voltage regulator shall provide regulated voltages to be utilized in all transistorized circuits within the receiver assembly.

3.4.1.4.3 Decoder Subassemblies - The decoder shall consist of five subassemblies each containing circuitry for decoding four audio code channels providing a total of 20 channels. Each of the fice decoder subassemblies shall be of plug-in type construction.

3.4.2 Base, Mounting MT-2740(*)/DRW-29 -

3.4.2.1 Function - The mounting base shall provide a means of mounting the Receiver, Radio R-1136(*)/DRW-29. The mounting base shall include a hand operated mechanism to permit rapid installation and removal of the receiver.

3.4.2.2 Form Factor - The maximum external dimensions of the mounting base, including vibration isolators and with the receiver mounting device fully extended, shall not exceed 2 by 6-1/2 by 9-13/16 inches.

3.4.2.3 Weight - The weight of the mounting base, including vibration isolators, shall not exceed one pound.

3.4.2.4 Vibration Isolators - Suitable vibration isolators shall be provided on the bottom surface of the mounting base.

4. QUALITY ASSURANCE PROVISIONS

4.1 General - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests

(2) Production Sample Tests

(3) Acceptance Tests

(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. This data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating conditions. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor's demonstration tests and when requested by the procuring activity, the equipment shall be delivered to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Fabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the current version before use.

4.2.5 Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by this government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first ten production equipments submitted for acceptance and sent to a designated government laboratory for tests. The equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defects made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test, the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commerical testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptable tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Manufacturing Reliability Test

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure equipment operation, in accordance with this specification.

4.4.1.3 Manufacturing Reliability Test - Each equipment shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency with the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's.

The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the six-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.

mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the six-hour period of operation following the ten-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The six-hour period specified above may be composed to two three-hour periods to conform with standard working hours.

4.4.2 Sampling Tests - Sampling tests shall be conducted on equipments which have successfully passed the individual tests. Equipments shall be selected for sampling tests by the Government Inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

4.4.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, production and special tests.

(3) Manufacturing reliability test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the currentversion before use.

of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling tests.

4.4.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows, except that when a Production Sample is requested and delivered to a government laboratory for test (See paragraph 4.3.) the requirement for a special test on an early production equipment may be omitted.

(1) On an early production equipment.

(2) On an early equipment after an engineering change.

(3) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the procuring activity.)

4.4.3.2 Scope of Tests - Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only those tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (60 minutes cycling in each of 3 planes)

(3) Humidity Tests

4.4.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the current version before use.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft)
Humidity	Room ambient
DC Voltage	27.0 ±2.0 volts

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to

Source: https://assist.dla.mil -- Downloaded: 202610-19T17:26Z
Check the source to verify that this is the current version before use.

permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor may resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the current version before use.

contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity.

4.7 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.8 Rejection and Retest - Equipment which has been rejected may be reworked and have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

4.9 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 *Intended Use* - The equipment covered by this specification shall be intended to be used as an airborne control radio receiver in pilotless aircraft and missiles.

6.2 Test Values - Normal and limiting values of performance data shall be determined at an input voltage of 27.5 ±0.5 V DC. This data is to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the current version before use.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.7. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

Source: https://assist.dla.mil -- Downloaded: 202640-19T17:26Z
Check the source to verify that this is the currentversion before use.

NOTICE - When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may be in any way related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:26Z
Check the source to verify that this is the current version before use.